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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID:

1. Topic (12 words max)

Water Quality Monitoring, Alerting and Pool Maintenance System

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Swimming pools must maintain high water quality to ensure public safety, but most facilities still rely on manual testing, fixed maintenance cycles, and human observation. These methods are delayed, error-prone, and unable to detect real-time risks such as water contamination, drowning incidents, or equipment failures. This creates risks, especially for children, the elderly, and immunocompromised users.

The problem affects pool managers, staff, and daily users. Socially, poor water conditions can lead to disease outbreaks or accidents. Technically, there is a lack of intelligent, automated systems that monitor and respond to changing conditions. Economically, inefficient chemical use and delayed maintenance increase operational costs. Environmentally, excess chemicals can leak into nearby water systems, causing ecological damage. Ethically, clean and safe water should be a basic standard in all public and private facilities.

This project aligns with the **Centre of Excellence in AI (CEAI)**, under the **Smart Environments, Public Safety, and Facility Management** verticals. It uses AI to transform monitoring and maintenance from reactive to intelligent — through behavior recognition, water classification, and predictive alerts.

It also supports key **Sustainable Development Goals (SDGs)**:

- **SDG 6**: Clean Water and Sanitation
- **SDG 3**: Good Health and Well-being
- **SDG 11**: Sustainable Cities and Communities

By addressing safety, sustainability, and operational efficiency, the problem presents a strong case for AI-driven innovation in modern pool management.

6. Brief description of Existing Research and Systems (300 words max)

Several commercial and open-source systems exist for basic pool water quality monitoring, such as **Blue Connect**, **pHin**, and **WaterGuru**. These products use IoT sensors to monitor pH, chlorine, and temperature, providing app-based feedback to users. However, they primarily function as **standalone monitoring tools** and lack predictive intelligence or real-time safety response.

Recent research has explored AI applications in environmental monitoring. Studies have applied **machine learning models** to predict water quality in rivers, lakes, and reservoirs using multi-parameter sensor data. Similarly, **computer vision** has been used to detect surface contamination and algae in water bodies. Some drowning detection prototypes use **CNN-based movement tracking** via surveillance cameras, but are rarely deployed in pool environments due to high complexity or cost.

Despite progress, most existing solutions have **key limitations**:

- They lack **AI-driven decision-making** or predictive maintenance
- Many systems only trigger alerts **after** thresholds are breached rather than anticipating issues
- Vision-based drowning detection remains **underdeveloped and not real-time** in many public settings
- Current products are often **expensive**, closed-source, and designed for individual consumers rather than shared facilities like public pools or hotels

In terms of research gaps, there is limited integration of **both sensor and vision data in a unified AI framework** for pool environments. There is also a need for **dynamic maintenance scheduling** based on usage and water trends — not just timer-based cleaning or manual planning.

This project aims to bridge these gaps by combining **machine learning**, **computer vision**, and **real-time alerting** into a single, intelligent system optimized for both safety and operational efficiency in pool management.

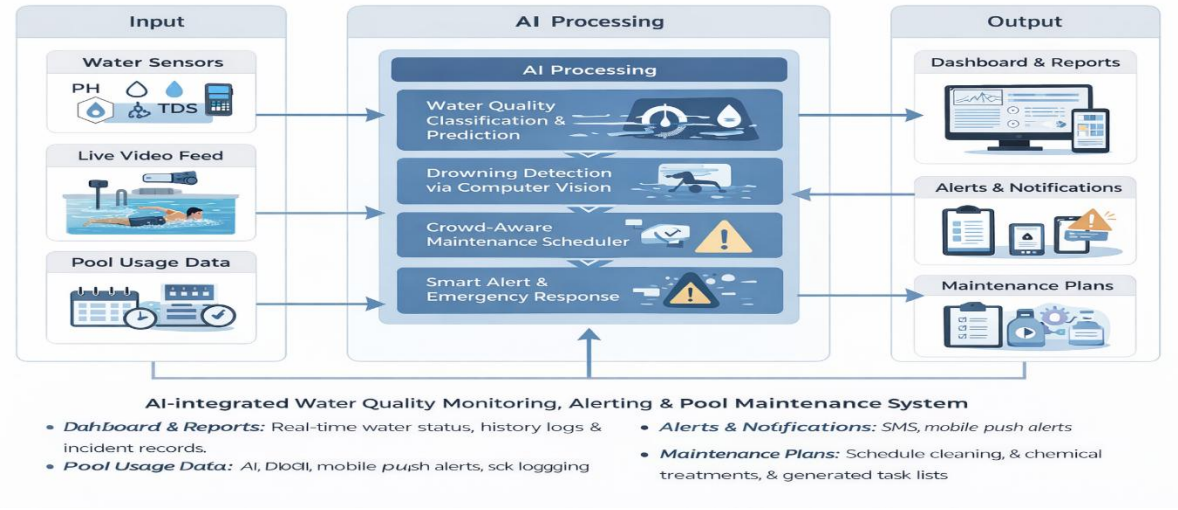
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

This project proposes an AI-integrated solution that overcomes the major limitations of traditional pool monitoring systems, such as delayed detection, lack of predictive capability, and isolated sensor use. Most current systems provide only basic alerts after thresholds are breached. In contrast, our system introduces real-time monitoring, behavior recognition, and predictive maintenance using both sensor data and computer vision. It focuses on improving pool safety, water quality, and operational efficiency through four key modules working together in one intelligent platform.

The first module, water quality classification and prediction, uses machine learning to process multi-parameter sensor data such as pH, turbidity, TDS, temperature, and chlorine. It classifies water as Safe, Warning, or Critical, and forecasts quality changes before they occur. The second module employs computer vision to analyze swimmer behavior via live camera feeds. Using CNN-based models, it detects early signs of distress, such as motionlessness or abnormal movements, and triggers real-time drowning alerts. The third module handles maintenance automation. By analyzing crowd usage history and water condition trends, it intelligently schedules cleaning and chemical treatment, avoiding unnecessary or delayed actions. The fourth module is an emergency-aware alert system that prioritizes incidents based on severity and notifies staff or users via SMS or mobile notifications.

The system's input includes live sensor values, camera footage, and pool usage data. These inputs are processed through AI algorithms for classification, behavior detection, forecasting, and task generation. The outputs are delivered through a dashboard, visual and mobile alerts, scheduled maintenance logs, and incident reports.

This comprehensive solution reduces human dependence, ensures continuous safety, and optimizes resource use in pool management. It aligns with the CEAI research cluster under Smart Environments and Public Safety, offering practical applications in schools, hotels, recreation centers, and municipal pools.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of this project requires interdisciplinary expertise spanning artificial intelligence, computer vision, sensor-based systems, and basic pool management principles. Essential domain expertise related to water quality standards, pool safety regulations, and maintenance practices can be gained through a review of industry guidelines, academic literature, and consultations with pool maintenance personnel or facility managers. Understanding parameters such as pH, chlorine levels, turbidity, and swimmer safety norms is crucial for defining meaningful thresholds and system behaviors.

The technical knowledge required for the project includes machine learning, deep learning, computer vision, and IoT integration. Team members can acquire and strengthen these skills through structured coursework, online learning platforms, and hands-on experimentation. Proficiency in programming languages such as Python is essential, along with familiarity with frameworks like TensorFlow or PyTorch for model development, OpenCV for image processing, and microcontroller platforms for sensor data acquisition. Knowledge of data preprocessing, model evaluation metrics, and real-time system integration will be developed during implementation and testing phases.

Data for the project will be collected from multiple sources. Water quality data will be obtained through real-time sensors measuring parameters such as pH, turbidity, temperature, and TDS. Pool usage and crowd data can be collected using camera-based monitoring and historical usage logs. For training and validation of AI models, publicly available datasets related to water quality and human activity recognition may be used, supplemented with locally collected data where feasible. Synthetic or simulated data may also be employed during early development to validate system performance.

Regarding ethical considerations, the project may require **ethical clearance**, particularly for the use of camera-based monitoring involving human subjects. To address this, the system will avoid facial recognition, anonymize video data, and restrict usage to behavior analysis only. Data will be stored securely, and appropriate consent and institutional guidelines will be followed to ensure privacy, transparency, and ethical compliance.

9. Objectives and Novelty

Main Objective To design and develop an **AI-based intelligent pool monitoring system** that ensures safe water quality, enhances swimmer safety, and optimizes pool maintenance by integrating machine learning, computer vision, real-time alerting, and predictive analytics.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22156310	Design a crowd-aware maintenance scheduling and optimization module.	• Analyze pool usage and crowd density data • Correlate crowd patterns with water quality degradation • Generate dynamic maintenance and chemical dosing schedules • Optimize resource usage and maintenance timing	Replaces fixed maintenance schedules with **AI-driven, usage-aware maintenance planning**, improving efficiency and sustainability.
IT22247254	Develop an AI-based water quality classification and prediction module.	• Collect and preprocess sensor data (pH, turbidity, TDS, temperature, chlorine) • Train ML models to classify water quality into Safe, Warning, and Critical • Implement time-series forecasting to predict future water quality trends • Integrate predictions with alert generation	Introduces **predictive water quality intelligence** instead of threshold-based monitoring, enabling early warnings before unsafe conditions occur.

IT22248480	Implement a computer vision–based swimmer behavior and drowning detection system.	• Capture and preprocess pool video streams. • Train CNN-based models to analyze swimmer movements pattern. • Detect abnormal behaviors such as prolonged inactivity or sinking. • Trigger real-time emergency alerts.	Provides **Early drowning detection using AI-based behavior analysis**, reducing dependence on manual lifeguard observation.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				